

KCA MAIN CAMPUS

PHAMARCY POS MANAGAEMENT SYSTEM

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Abstract

A Point of Sale (POS) system plays a critical role in modern pharmacy operations by facilitating efficient sales transactions, medicine inventory management, prescription tracking, and report generation. Many community pharmacies and retail drug stores in Kenya still rely on manual or semi-automated systems that are prone to errors, data loss, regulatory non-compliance, and inefficiency. This project proposes the design and implementation of a computerized Pharmacy POS Management System aimed at improving accuracy, speed, and reliability in pharmacy sales operations.

The proposed system will allow pharmacy staff to record sales transactions, manage medicine inventory with expiry date tracking, generate receipts, process prescriptions, and produce sales and stock reports in real time. The system will be developed using modern software development tools and methodologies to ensure reliability, usability, and scalability. The Waterfall development model will be adopted to allow systematic planning, development, and testing throughout the development process. The expected outcome of this project is a user-friendly and efficient Pharmacy POS system that enhances pharmacy operations, reduces human errors, ensures regulatory compliance, and supports informed decision-making through accurate reporting.

Declaration

I declare that this project proposal is my original work and has not been submitted to any other institution for academic credit. All sources of information used have been duly acknowledged.

Student Name: _____

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Acknowledgment

I would like to express my sincere gratitude to my supervisor for the guidance, support, and encouragement provided throughout the preparation of this project proposal. I also acknowledge the support from my lecturers, classmates, friends, and family for their valuable input and motivation during the development of this work.

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Chapter One: Introduction

1.1 Background

The rapid advancement of information technology has significantly transformed healthcare operations across various sectors, including retail pharmacy services. One of the key areas affected by these advancements is medicine dispensing and sales transaction management. A Pharmacy Point of Sale (POS) system is a computerized solution that enables pharmacies to process sales transactions, manage medicine inventory, track expiry dates, handle prescriptions, and generate reports efficiently. In today's competitive healthcare environment, accurate and fast transaction processing is essential for customer satisfaction, patient safety, and effective pharmacy management.

MedPlus Pharmacy, a community pharmacy located in Nairobi, Kenya, serves as the case study for this project. MedPlus Pharmacy is a small-to-medium enterprise (SME) that operates as a retail drug store serving the local community. The pharmacy stocks over 500 different pharmaceutical products including prescription medicines, over the counter (OTC) drugs, medical supplies, and health supplements. Currently, MedPlus Pharmacy relies on manual record-keeping using paper-based registers and basic spreadsheets for inventory tracking. This has led to several operational challenges including:

- Difficulty tracking medicine expiry dates
- Inaccurate stock levels leading to stockouts or overstocking
- Slow transaction processing during peak hours
- Errors in calculating prices and discounts
- Challenges in generating sales and inventory reports for management
- Risk of dispensing errors affecting patient safety

The adoption of a modern Pharmacy POS system will help MedPlus Pharmacy streamline operations, minimize losses from expired medicines, ensure regulatory compliance with the Pharmacy and Poisons Board of Kenya requirements, and enhance decision-making. This project focuses on designing and implementing a Pharmacy POS Management System tailored to meet the specific needs of community pharmacies like MedPlus Pharmacy.

1.2 Problem Statement

Community pharmacies like MedPlus Pharmacy face significant challenges due to reliance on manual and semi-automated sales systems. These systems are prone to errors such as incorrect pricing, poor medicine inventory tracking, failure to monitor expiry dates, delayed report generation, and difficulty managing prescriptions. These challenges result in financial losses from expired stock, poor customer service, regulatory compliance issues, and inefficient pharmacy operations. There is a need for a reliable and automated Pharmacy POS Management System that can accurately manage sales transactions, medicine inventory with expiry tracking, prescription records, and reporting in real time.

1.3 Objectives

1.3.1 Main Objective

To design and implement an efficient Pharmacy POS Management System for MedPlus Pharmacy that automates sales transactions, medicine inventory management, and prescription tracking.

1.3.2 Specific Objectives

To develop a system that records and processes pharmacy sales transactions accurately, including prescription and OTC medicines.

To implement a medicine inventory management module that updates stock levels in real time and tracks expiry dates with automated alerts.

To create a prescription management module that records prescription details and links them to sales transactions.

To generate automated sales, inventory, and expiry reports for pharmacy management decision-making.

To design a user-friendly interface that enhances ease of use and efficiency for pharmacy staff.

1.4 Justification

The proposed Pharmacy POS Management System will improve efficiency, accuracy, and accountability in pharmacy transactions. MedPlus Pharmacy owners and staff will benefit from faster transaction processing, reliable medicine stock tracking, expiry date alerts, and comprehensive reports. The system will help the pharmacy comply with Pharmacy and Poisons Board regulations by maintaining proper records of prescription sales. The project introduces a modern and scalable solution suitable for community pharmacies and retail drug stores, contributing to improved healthcare service delivery and IT-based pharmacy management practices in Kenya.

1.5 Scope and Limitations

Scope:

The scope of this project includes:

- Sales processing for prescription and OTC medicines
- Medicine inventory management with expiry date tracking
- Prescription record management
- Receipt generation
- Low stock and expiry alerts
- Sales and inventory report generation
- User management with role-based access (Admin, Pharmacist, Cashier)

The system will be designed specifically for MedPlus Pharmacy, a community pharmacy in Nairobi, Kenya, but can be adapted for similar retail pharmacies.

Limitations:

- Time constraints within the academic calendar
- Limited financial resources for hardware acquisition
- Dependency on available software tools
- The system will not integrate with external systems such as NHIF or supplier databases in this initial version
- The system will not include controlled substance tracking features in this phase

Chapter Two: Literature Review

2.1 Introduction

The literature review examines existing Pharmacy Point of Sale (POS) systems, related technologies, and research works to understand how pharmacy sales transaction and inventory management systems have evolved over time. This chapter is important because it provides a foundation upon which the proposed Pharmacy POS Management System is designed. By reviewing existing pharmacy POS solutions used in real-life pharmacy environments, this study identifies strengths, weaknesses, and gaps that inform the design decisions of the proposed system. The review focuses on commercial pharmacy POS systems, open-source solutions, and academic studies related to pharmacy POS systems, medicine inventory management, prescription tracking, and transaction processing.

2.2 Review of Related Works

2.2.1 Real-Life Commercial Pharmacy POS Systems

Several POS systems are currently used by pharmacies worldwide to manage sales, medicine inventory, prescriptions, and reporting. These systems demonstrate how POS technology is applied in real-world pharmacy scenarios.

McKesson Pharmacy Systems is a widely used pharmacy management solution in North America. It provides features such as prescription processing, sales management, inventory control, drug interaction checking, and compliance reporting. While comprehensive, McKesson

is expensive and designed for large pharmacy chains, making it unsuitable for small community pharmacies in developing countries like Kenya.

Liberty Software Pharmacy System offers point of sale, prescription management, inventory control, and patient management features. It integrates with various pharmacy peripherals including barcode scanners and receipt printers. However, it requires significant upfront investment and ongoing subscription costs.

mPharma is an African-based pharmacy management platform that helps pharmacies manage inventory and provides access to affordable medicines. While mPharma focuses on supply chain optimization, it provides useful insights into pharmacy management challenges in African contexts. However, it operates on a specific business model that may not suit all independent pharmacies.

RxNova POS is a pharmacy-specific point of sale system that handles both prescription and OTC sales, manages inventory, and generates reports. It is designed for retail pharmacies but requires licensing fees that may be prohibitive for small pharmacies.

These commercial systems demonstrate that while pharmacy POS solutions are effective, many are costly, complex, or designed for large organizations or specific markets, creating a gap for affordable and customizable systems tailored to smaller community pharmacies in Kenya.

2.2.2 Open-Source and Custom Pharmacy Systems

Open-source pharmacy management systems such as OpenMRS and Bahmni provide customizable alternatives for healthcare facilities. While primarily designed for hospital settings, these systems include pharmacy modules that manage drug dispensing and inventory. However, they are often too complex for standalone retail pharmacies and require significant technical expertise for customization and deployment.

Odoo POS with Pharmacy Modules offers a customizable open-source solution that can be configured for pharmacy operations. It provides inventory management, sales processing, and reporting features. Although Odoo reduces licensing costs, it requires technical skills for customization, deployment, and maintenance specific to pharmacy requirements.

Custom-built pharmacy POS systems developed for specific pharmacies are common in Kenya, especially in small retail drug stores. These systems are tailored to business needs but may lack scalability, proper documentation, expiry tracking features, or long-term support. Poor system design can lead to data inconsistency and security risks.

2.2.3 Technology Overview

Modern pharmacy POS web applications are built using web technologies that support scalability, security, and usability. Programming languages such as Python, PHP, Java, and JavaScript are widely used. Frameworks like Django, Laravel, and React provide structured approaches to building secure and maintainable web applications. Databases such as MySQL, PostgreSQL, and SQLite are commonly used to store transactions, medicine inventory, prescription records, and user data.

Most pharmacy POS web applications follow a client-server architecture, where users interact with the system through a browser, the application logic processes transactions on the server, and the database stores persistent data. This architecture supports centralized data management, real-time updates, and role-based access control essential for pharmacy operations.

Key features required in pharmacy POS systems include:

- Medicine inventory management with batch and expiry tracking
- Barcode scanning for fast product identification
- Prescription recording and validation
- Price management with discount capabilities
- Receipt generation with pharmacy details
- Reporting for sales, inventory, and expiry analysis
- User authentication with role-based access

2.2.4 Research Gaps

Despite the availability of numerous pharmacy management systems, several gaps remain:

1. Cost Barrier: Many systems are expensive, requiring subscription fees or significant upfront investments that small community pharmacies in Kenya cannot afford.

2. Complexity: Enterprise systems are designed for large pharmacy chains and include features unnecessary for small pharmacies, making them difficult to use and maintain.
3. Context Specificity: Most commercial systems are designed for Western markets and may not address specific requirements of Kenyan pharmacies, such as local regulatory compliance and common payment methods.
4. Expiry Management: Some basic POS systems provide limited expiry date tracking and alerting features, which is critical for pharmacy operations to prevent dispensing expired medicines.
5. Integration: There is a lack of affordable, standalone pharmacy POS systems that can operate independently without requiring complex integrations.
6. Academic Focus: There is a lack of customizable pharmacy POS systems designed specifically for academic learning and demonstration purposes that can also serve real community pharmacies.

These gaps justify the development of the proposed Pharmacy POS Management System for MedPlus Pharmacy.

2.3 Theoretical Framework

This project is guided by the System Development Life Cycle (SDLC) theory, which provides a structured approach to software development. The SDLC model includes phases such as requirement analysis, system design, implementation, testing, deployment, and maintenance. Applying SDLC ensures that the Pharmacy POS Management System is systematically planned, developed, tested, and evaluated.

In addition, principles of Database Management Systems (DBMS) guide the data design to ensure data integrity, consistency, and efficient storage of medicine inventory, sales transactions, and prescription records.

The Technology Acceptance Model (TAM) informs the user interface design, emphasizing perceived usefulness and ease of use to ensure pharmacy staff readily adopt the new system.

Furthermore, principles of inventory management theory, including First-In-First-Out (FIFO) and safety stock concepts, guide the design of the inventory management module to ensure proper medicine stock rotation based on expiry dates.

2.4 Summary

This chapter reviewed real-life commercial pharmacy POS systems, open-source solutions, and supporting technologies. The review highlighted the strengths and limitations of existing pharmacy systems and identified gaps related to cost, complexity, context specificity, and expiry management. These findings inform the design of the proposed Pharmacy POS Management System, which aims to provide an affordable, efficient, and user-friendly system tailored to community pharmacies like MedPlus Pharmacy in Nairobi, Kenya.

Chapter Three: Methodology

3.1 Introduction

This chapter describes the methodology that will be used to design, develop, test, and deploy the proposed Pharmacy POS Management System for MedPlus Pharmacy. The methodology provides a structured approach that guides the entire system development process, ensuring that the final system meets user requirements, is reliable, and is easy to maintain. The chapter discusses the system design, tools and technologies, data requirements, development approach, testing strategies, and project timeline.

3.2 System Design

System design defines how the Pharmacy POS Management System will be structured and how its components will interact. Proper system design is essential to ensure efficiency, security, and scalability.

3.2.1 System Architecture

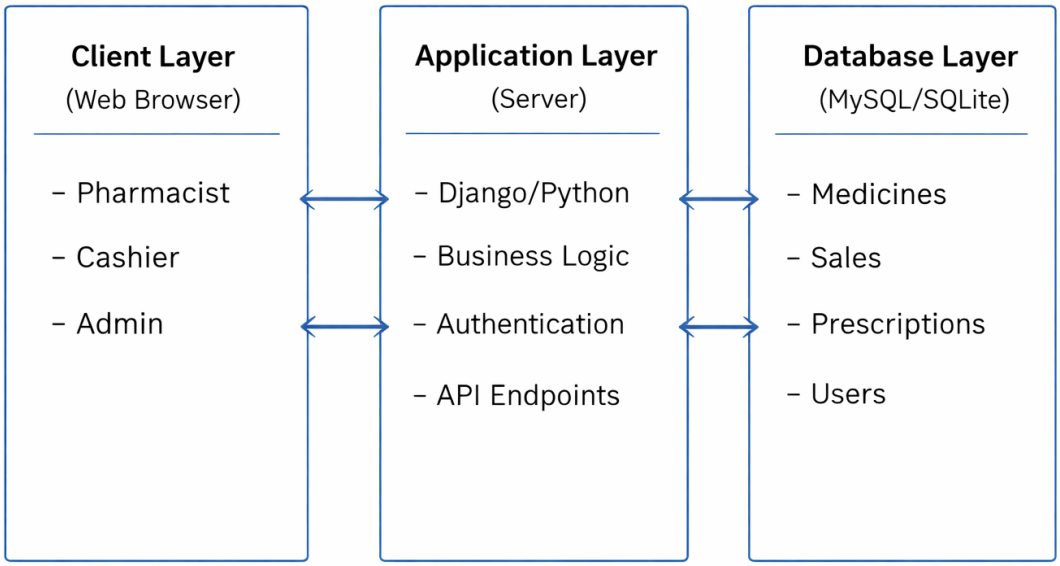
The proposed Pharmacy POS system will adopt a client-server architecture. In this architecture:

- Client Layer: Users (Pharmacists, Cashiers, Admin) interact with the system through a web browser, which serves as the client.

- Application Layer: The application logic resides on the server, where requests are processed, business rules are applied, and data is managed.
- Database Layer: The database stores all persistent data such as medicines, sales transactions, prescriptions, users, and inventory records.

This architecture allows centralized data management, improves security through controlled access, and supports multiple users accessing the system simultaneously within the pharmacy environment.

System Architecture Diagram:



3.2.2 System Modules / Components

The Pharmacy POS Management System will be divided into the following major modules:

1. User Authentication and Authorization Module

- Manages user login, logout, and session management
- Role-based access control (Admin, Pharmacist, Cashier)
- Password management and security

2. Medicine Inventory Management Module
    - Manages medicine records (name, category, manufacturer, price, quantity)
    - Batch and expiry date tracking
    - Stock level updates (additions, deductions)
    - Low stock alerts and notifications
    - Expiry alerts for medicines approaching expiration
    - Medicine categories (Prescription, OTC, Supplements, Medical Supplies)
3. Sales Management Module
    - Handles sales transactions for prescription and OTC medicines
    - Item selection via search or barcode scanning
    - Pricing with discount capabilities
    - Multiple payment methods (Cash, M-Pesa, Card)
    - Receipt generation and printing
    - Sales history and transaction lookup
4. Prescription Management Module
    - Records prescription details (prescriber, patient, date)
    - Links prescriptions to sales transactions
    - Maintains prescription history for compliance
5. Reporting Module
    - Daily, weekly, and monthly sales reports
    - Inventory status reports
    - Expiring medicines reports
    - Low stock reports
    - Profit margin analysis
    - Export reports to PDF/Excel
6. User Management Module
    - Add, update, and deactivate system users

- Assign roles and permissions
- View user activity logs

3.3 Tools and Technologies

The development of the Pharmacy POS Management System will utilize modern and reliable technologies as follows:

Component	Technology	Justification
Programming Language	Python 3.x	Simplicity, readability, strong community support, extensive libraries
Web Framework	Django 4.x	Built-in security features, authentication mechanisms, admin interface, rapid development
Frontend	HTML5, CSS3, Bootstrap 5, JavaScript	Responsive design, cross-browser compatibility, interactive interfaces
Database (Development)	SQLite	Lightweight, no separate server required, easy setup
Database (Production)	MySQL/PostgreSQL	Robust, scalable, reliable for production environments

IDE	Visual Studio Code	Lightweight, extensions support, Git integration
Version Control	Git/GitHub	Source code management, collaboration, backup
Barcode Library	python-barcode	Generate and read barcodes for medicines
PDF Generation	ReportLab / WeasyPrint	Generate receipts and reports in PDF format
Charts/Visualization	Chart.js	Interactive charts for dashboard and reports

Hardware Requirements for MedPlus Pharmacy:

Hardware	Description	Purpose
Computer/Laptop	Intel Core i3+, 4GB RAM, 256GB Storage	Run the POS application
Barcode Scanner	USB handheld scanner	Scan medicine barcodes
Receipt Printer	80mm thermal printer	Print sales receipts
Cash Drawer	Electronic cash drawer	Secure cash storage

3.4 Data Collection and Requirements

3.4.1 Data Requirements

The system will require various types of data, including:

Medicine Data:

- Medicine ID/Barcode

- Medicine name (generic and brand)
- Category (Prescription, OTC, Supplement, Medical Supply)
- Manufacturer
- Unit price (buying and selling)
- Quantity in stock
- Batch number
- Expiry date
- Reorder level

Sales Transaction Data:

- Transaction ID
- Date and time
- Items sold (medicine, quantity, unit price, subtotal)
- Total amount
- Discount applied
- Payment method
- Cashier/Pharmacist ID
- Prescription reference (if applicable)

Prescription Data:

- Prescription ID
- Patient name
- Prescriber name and facility
- Date issued
- Medicines prescribed
- Linked transaction ID

User Data:

- User ID
- Username
- Password (hashed)

- Full name
- Role (Admin, Pharmacist, Cashier)
- Contact information
- Status (Active/Inactive)

3.4.2 Data Sources

Data will be obtained through:

- User input: Administrators will enter medicine data, user information, and initial stock levels through system forms
- Transactions: Sales data will be captured in real-time during pharmacy operations
- Prescriptions: Prescription details will be entered by pharmacists during prescription sales
- Barcode scanning: Medicine identification during sales transactions

3.4.3 Data Collection Methods

Data will be collected using:

- Structured input forms within the Pharmacy POS application
- Barcode scanning for quick product identification
- Validation rules to ensure accuracy and consistency of data entered into the system
- Dropdown selections for standardized data entry (categories, payment methods, roles)

3.5 Development Approach

3.5.1 Development Methodology

The project will adopt the Waterfall Development Model. The Waterfall model is a linear and sequential software development methodology where each phase must be completed before the next phase begins. This model is selected for this academic project because:

- It is easy to understand and well-structured
- It allows clear documentation at each stage
- Requirements are well-defined upfront
- It provides predictable timelines and deliverables
- It is suitable for projects with fixed requirements

3.5.2 Waterfall Development Phases

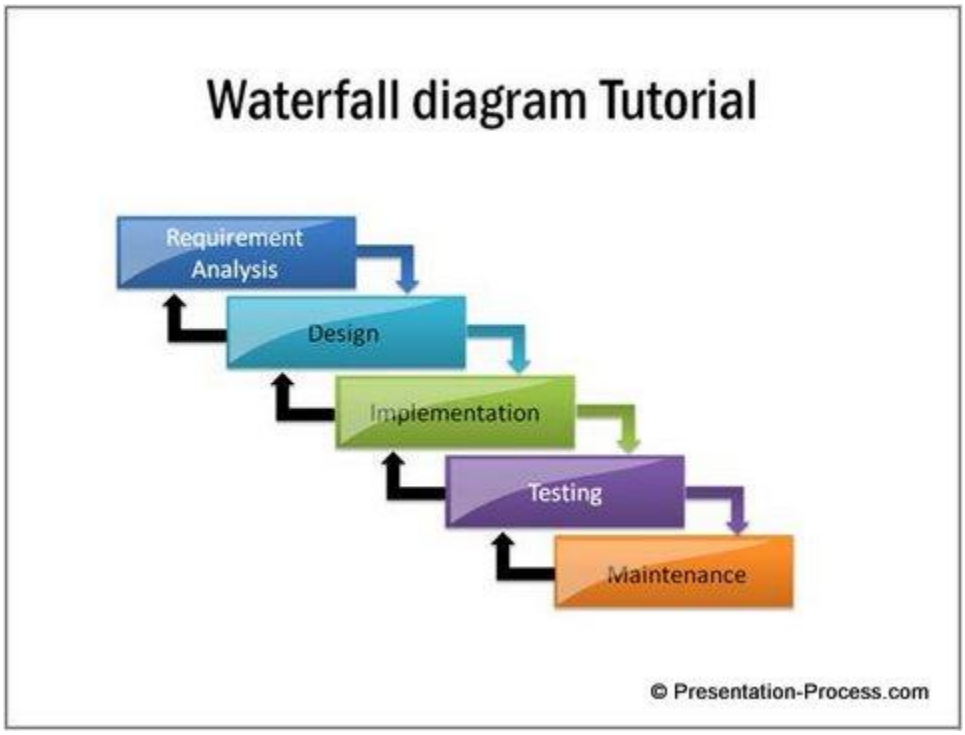
The Waterfall model for the proposed Pharmacy POS Management System consists of the following phases:

1. Requirements Analysis
    - Conduct interviews with MedPlus Pharmacy staff
    - Document functional and non-functional requirements
    - Identify user roles and use cases
    - Define system constraints and assumptions
2. System Design
    - Design system architecture (client-server)
    - Create database schema (ER diagrams)
    - Design user interface wireframes and mockups
    - Define module interactions and data flow
3. Implementation
    - Set up development environment
    - Develop database models in Django
    - Implement backend logic for each module
    - Create frontend templates and interfaces
    - Integrate barcode scanning and receipt printing
4. Testing

- Unit testing of individual modules
- Integration testing of module interactions
- System testing of complete application
- User acceptance testing with MedPlus Pharmacy staff

5. Deployment
    - Install and configure system at MedPlus Pharmacy
    - Migrate initial data (medicines, users)
    - Train pharmacy staff on system usage
    - Go-live and monitor system performance
6. Maintenance
    - Address bugs and issues reported by users
    - Implement minor enhancements
    - Provide ongoing support and documentation

3.5.3 Waterfall Diagram



3.6 Testing and Validation

Testing is essential to ensure that the Pharmacy POS Management System performs as expected and meets user requirements.

Testing Levels:

Testing Type	Description	Focus Areas
Unit Testing	Test individual modules separately	Medicine CRUD operations, sales calculations, user authentication
Integration Testing	Test interaction between modules	Sales updating inventory, prescription linking to transactions
System Testing	Test complete application	End-to-end workflows, performance, security
User Acceptance Testing (UAT)	Test with MedPlus Pharmacy staff	Usability, business requirements, real-world scenarios

Test Cases Examples:

Test Case	Input	Expected Output	Module
Add new medicine	Medicine details	Medicine saved, confirmation message	Inventory
Process sale	Items, payment	Sale recorded, receipt generated, stock updated	Sales
Login with valid credentials	Username, password	Dashboard displayed	Authentication
Generate expiry report	Date range	List of expiring medicines	Reporting

Low stock alert	Stock below reorder level	Alert notification displayed	Inventory

3.7 Project Timeline and Budget

Project Timeline (Gantt Chart):

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Requirements Analysis	[redacted]	[redacted]				
System Design		[redacted]	[redacted]			
Database Design			[redacted]	[redacted]		
Implementation (Coding)				[redacted]	[redacted]	[redacted]
Interface Development				[redacted]	[redacted]	
Testing & Validation					[redacted]	[redacted]
Deployment						[redacted]
Documentation	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]

Project Budget:

Category	Description	Estimated Cost (KES)	Estimated Cost (USD)

Software	Development tools (VS Code, Python, Django - Open Source)	0	0
Database	MySQL/SQLite (Open Source)	0	0
Hardware	Computer for development (existing)	0	0
Hardware (Pharmacy)	Barcode scanner	5,000	40
Hardware (Pharmacy)	Thermal receipt printer	15,000	115
Hardware (Pharmacy)	Cash drawer	8,000	60
Internet	Internet costs (6 weeks)	6,000	45
Stationery	Printing, binding	3,000	23
Transport	Site visits to MedPlus Pharmacy	4,000	30
Contingency	Miscellaneous expenses	5,000	40
Total		46,000	353

Note: Most development tools are open source, minimizing software costs. Hardware costs are estimates for pharmacy deployment.

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